

Does Structured Agent Memory Improve Prediction of Individual Dogs’ Choices?

A Retrospective Exploratory Evaluation of Hindsight

Author information pending

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Abstract

Individual behavioral history may support prediction without requiring a specialized memory system. We evaluated whether an implemented Hindsight retain/recall pipeline improves a language-model predictor’s probabilities for dogs’ laboratory cup choices beyond simpler ways of supplying the same observations. Public ManyDogs1 data yielded 420 eligible dogs; a deterministic subset of 16 dogs from 12 sites supplied eight earlier pointing trials and eight held-out trials in the other pointing condition per dog. Five arms used no individual history, full supplied history, a deterministic count profile, cosine top-three retrieval, or Hindsight recall. Fresh predictor contexts generated 132 query-specific probability vectors, reused across 128 held-out outcomes per arm. Hindsight supplied all eight historical trials whereas top-three retrieval supplied three, confounding evidence coverage with pipeline choice. Predictions were validated and hashed before scoring. Mean summed multiclass Brier scores were 0.5532, 0.3806, 0.3716, 0.4339, and 0.3761, respectively. Hindsight’s paired improvement over full history was 0.0046 (fixed-output dog-cluster 95% bootstrap interval $[-0.0087, 0.0169]$), and over the profile was -0.0044 ($[-0.0111, 0.0023]$). Its improvement over top-three retrieval was 0.0578 ($[0.0118, 0.1067]$), but the lower bound did not exceed an investigator-chosen practical threshold of 0.03. A separate 420-dog frequency-model diagnostic found that point-relative history worsened the primary score whereas physical-side history improved a secondary score. Training-only quality assurance corrected initial extraction failures before retrieval-arm prediction, making this an exploratory analysis with documented amendments. The results do not establish practical superiority of this Hindsight configuration over simple history inputs, prospective everyday behavior prediction, or benefit for any specified individual dog.

1 Introduction

Predicting a particular animal’s future response is different from describing average performance in a behavioral experiment. An animal’s earlier choices can provide useful information, but the predictive value depends on the response definition, context, chronology, representation of history, and comparison method. A specialized agent-memory system should therefore be evaluated against strong, inexpensive ways of supplying exactly the same historical evidence.

The motivating question concerned whether Hindsight could improve prediction of one dog’s everyday behavior. Available public data did not support a prospective test of that claim. We instead constructed an explicitly narrower, retrospective laboratory proxy: prediction of the same dog’s left-cup, right-cup, or no-choice response after a pointing gesture, using a frozen earlier block

of observations. The unit of generalization is a dog-level task drawn from an eligible public cohort; it is not a designated pet followed into the future.

Our principal contribution is an auditable comparison of an actual local Hindsight implementation with four controls, accompanied by source-to-memory fidelity checks and independent scoring reconstruction. A larger, separate statistical diagnostic examines whether basic representations of individual history are informative at all. We preserve its unfavorable primary result rather than substituting a favorable secondary result. Neither analysis measures canine mental states or constructs a replica of a dog’s cognition.

2 Related work and scope

ManyDogs1 investigated dogs’ responses to human pointing using coordinated procedures across research sites [1, 2]. We reuse its trial annotations for a new predictive comparison; the source researchers did not report the memory-system results presented here. Ostensive and nonostensive pointing differ in communicative cues, providing an observed change of experimental condition. This does not establish that the second style was unfamiliar throughout the dog’s life.

Hindsight proposes structured memory operations for language-model agents [4]. Our evaluation uses the open implementation’s *retain* and *recall* operations, with a particular local extraction model and retrieval configuration [5]. We do not evaluate its *reflect* operation, all supported configurations, or general long-term-memory benchmarks. Conventional vector retrieval is implemented on the same expanded observation corpus and embedder. Full history and a deterministic count profile are essential controls because the entire supplied history contains only eight observations.

Probability forecasts are evaluated primarily using the Brier score [6]. Classification accuracy alone cannot distinguish a cautiously correct forecast from an unjustifiably confident one. We report accuracy and log loss as secondary summaries, and treat calibration and explanation audits as descriptive measurements.

3 Data and experimental design

3.1 Source data, eligibility, and response

The public main CSV contains records for 704 dogs. Eligibility required the source authors’ consistently Included status; interpretable and consistent condition order; sequential experimental attempt numbering; source rows consistent with the declared block order; and at least eight attempted trials in each condition with known physical pointed side and response among $\{L, R, NC\}$ in the first eight attempts. We did not skip an early unknown outcome to choose a later convenient row. This produced 420 dogs across 20 sites and excluded 284 dogs, sometimes for multiple reasons (Appendix A). Completion and label-availability selection limit applicability to the original source population.

The endpoint is the source-coded first physical contact with either cup using the snout or a front paw within 25 seconds after release: *L* for the physical left cup, *R* for the physical right cup, and *NC* when neither cup is chosen. We used existing annotations, without independently re-scoring videos. Letters encode positions, not correct/incorrect choices. No-choice attempts were retained, including those whose source `valid_trial` field is unavailable; that field and completion status were not predictor inputs. This endpoint differs from the originally proposed approach/withdraw/orient/no-response everyday-behavior codebook.

3.2 Chronological split and deterministic subset

For each dog, the first eight attempted trials of the first pointing condition formed the history, and the first eight attempted trials of the second condition formed the test block. Ostensive pointing includes eye contact and speaking the dog’s name; nonostensive pointing uses throat clearing and averted gaze. Pointed cup side is observable before response. Warmup, odor-control, and other outcomes were not supplied. Some additional first-condition attempts occurred after the supplied eight and before the test block. These were omitted identically across arms: 100 attempts for 73 dogs in the 420-dog diagnostic, and eight attempts for five dogs in the 16-dog comparison. This is a selected historical snapshot, not complete experience through the prediction time.

The language-model comparison selected the first 16 eligible subjects after sorting the SHA256 digest of the UTF-8 string 20260930: +source subject identifier. Selection did not use choice values beyond eligibility for observed labels. Neutral identifiers D01–D16 replaced source identities in predictor packets. Sites, demographics, source URLs, and publication names were omitted to reduce lookup opportunities. These 16 dogs represented 12 sites and contributed 128 held-out outcomes: 68 left choices, 57 right choices, and three no choices. The fixed sample size was a feasibility choice, without a prospective power calculation.

Each dog had only two distinct current inputs: the second pointing condition with a point to *L* or *R*. Future attempt number and schedule were omitted. With memory frozen and no outcome updates, one probability vector per distinct input was reused across matching held-out trials. The no-history arm required only four style-by-side inputs reused across dogs. Therefore 128 scored outcomes per arm do not mean 128 independent model generations.

3.3 Five evidence arms

All arms shared task definitions and prediction instructions, and differed in supplied past observations:

- (1) **No history:** current pointing condition and side only.
- (2) **Full history:** all eight raw historical trial records, with explicit physical-side definitions. “Full” refers to the supplied eight observations.
- (3) **Profile:** deterministic counts of physical choices and cue-following, including counts conditional on pointed side. No anecdotes were selected by the researcher.
- (4) **Vector retrieval:** cosine top-three expanded event texts using normalized 384-dimensional `intfloat/multilingual-e5-small` ONNX embeddings, mean pooling, and the embedder’s query/passage prefixes. The pinned revision is recorded in Appendix C. Equal scores were broken by more recent attempt first.
- (5) **Hindsight:** actual retained per-dog banks and recalled text from Hindsight 0.10.2. Retention used local Ollama `qwen3.5:4b`; recall used high budget, a 4,096-token cap, and no raw chunks. The same pinned embedder served this arm. RRF reranking was configured; observation consolidation was disabled.

The eight source observations and current questions were the same across history arms. Their representation and quantity returned to the predictor differed: top-three retrieval deliberately exposed three events, while audited Hindsight recall effectively exposed all eight. Thus the Hindsight–retrieval contrast is not an isolated test of a memory architecture at matched evidence count. The token cap is a maximum, not an equalized realized input length.

3.4 Predictors, separation, and prediction lock

Each dog-arm task used a fresh Codex subagent context with no inherited conversation; that task generated the two side-specific probability vectors and brief rationales. A separate fresh context generated the four no-history vectors. Overall, 65 output files contain 132 unique vectors. Predictors inherited the same configured model setting and common prompt, but the interface did not expose the exact underlying model revision, temperature, seed, token usage, or pure inference time. We did not repeat generations to estimate stochastic variability.

Predictors were instructed to read only their assigned packet, not browse, not read other files, and not consult other predictors. A curator prepared the chronological source split and separate test-label file. The predictor coordinator did not inspect test outcomes or tune predictions; it once inspected a profile packet’s schema without passing its content to fresh child predictors. The workspace was shared: separation was enforced by instructions and recorded workflow, not a hardened access-control boundary. This qualification matters because hashes cannot prove historical isolation.

Outputs required nonnegative finite probabilities $[p_L, p_R, p_{NC}]$, an abstention flag, and a brief evidence-based rationale. The scorer validated dimensions, identifiers, and a sum within 0.001 of one, then normalized by the sum. The local plan allowed one format-only retry followed by a development-distribution fallback. No retry or fallback was needed. All 65 valid files were SHA256-locked, together with plan and scorer hashes, before the scorer opened the held-out labels. Predictions were not regenerated after seeing results.

3.5 Training-only quality-assurance amendment

Initial retention sometimes interpreted L/R as incorrect/correct and omitted or contradicted source trials. These failures were detected against training records before any Hindsight or vector-retrieval predictor started and before held-out scoring. Some full-history and profile generations had already started. Both retrieval arms then received deterministic expanded descriptions such as “the human pointed to the physical left cup; the dog chose the physical right cup,” together with explicit physical-code definitions. Hindsight’s retention mission requested independent preservation of every attempt and stated that behavioral dates were unknown.

A separate two-dog development check, selected by SHA256 of **retain-dev**: + pilot subject identifier, recovered all 16 source trial triplets before revised main retention began. The development cases did not include no-choice responses. New bank identifiers and versioned artifacts preserved the faulty initial extraction separately. The scored arm is the corrected pipeline; the initial one was not silently replaced after observing prediction performance. This documented local amendment, public-data exposure, and retrospective design mean the study is exploratory. The files named **prereg_*.md** are local analysis plans, not an external registry entry or an untouched confirmatory preregistration.

4 Metrics and inference

For event i , method a , and observed category y_i , the primary summed multiclass Brier score is

$$\text{BS}_{ia} = \sum_{k \in \{L, R, NC\}} (p_{iak} - \mathbb{I}\{y_i = k\})^2, \quad 0 \leq \text{BS}_{ia} \leq 2. \quad (1)$$

Scores are averaged across all attempted test outcomes, including abstentions. Each dog has eight test events, so event and dog weights coincide. Positive paired improvement favors Hindsight:

$$\Delta_a = \frac{1}{16} \sum_{d=1}^{16} (\overline{\text{BS}}_{da} - \overline{\text{BS}}_{dH}). \quad (2)$$

Accuracy uses deterministic argmax ties in L, R, NC order. Natural-log loss floors probabilities at 10^{-6} and then renormalizes. We also report coverage, wrong argmax predictions with maximum probability at least 0.8, and descriptive top-label expected calibration error in four fixed equal-width bins. With only three test no-choice outcomes, class-level calibration conclusions are especially unstable.

Paired percentile bootstrap intervals use 10,000 draws and seed 20260930. The primary resampling unit is the whole dog, preserving dependent trials and pairing across arms. A separate site sensitivity resamples whole sites with replacement and averages scores over all included events; sites with more sampled dogs receive more event weight. This sensitivity does not resolve all site dependence with only 12 sites. Neither resampling scheme reruns extraction or prediction: uncertainty is conditional on the realized fixed model outputs, public cohort, and selected implementation. Model-generation uncertainty, pipeline tuning, and public-data contamination are outside these intervals.

The investigator-chosen practical threshold was 0.03 summed Brier units. It is a decision convention, not an established canine-behavior standard or a validated utility difference. Practical benefit against an arm required both dog and site 95% lower bounds to exceed 0.03. To claim superiority over all three history baselines, the local rule required the minimum dog/site one-sided bootstrap lower quantile at 0.05/3 to exceed 0.03 for each. This Bonferroni-style rule is reported as specified; it does not convert an exploratory, small-sample analysis into confirmatory proof.

4.1 Separate statistical diagnostic

The 420-dog analysis used no language-model memory system. Its cohort contains the 16 language-model dogs: the two analyses are overlapping diagnostics, not independent replications. Pilot data for 61 separate dogs selected smoothing constants from $\{1, 2, 4, 8, 16, 32\}$. Other dogs' first-block histories formed leave-one-dog-out pooled frequencies, with add-one smoothing. The primary comparator conditioned pooled physical choices on pointed side. The primary history model estimated follow/opposite/no-choice counts, smoothed toward a uniform prior with total mass $\alpha = 2$, and mapped them to current physical side.

Secondary models used physical-side counts $\mathbf{c}_d$: uniform shrinkage $(\mathbf{c}_d + \alpha \mathbf{1}/3)/(8 + \alpha)$ with $\alpha = 4$, or pooled-side shrinkage $(\mathbf{c}_d + \alpha \mathbf{q}_{-d})/(8 + \alpha)$ with $\alpha = 8$. No second-block labels entered fitted predictors. Cluster intervals condition on those fitted probabilities rather than refitting pooled distributions within each resample. This diagnostic is a separate analysis of representations of history, not a larger Hindsight trial; its primary and secondary statuses were kept distinct.

5 Results

5.1 Language-model comparison

Table 1 reports all five arms. The profile achieved the lowest Brier and log-loss point estimates; full history had the highest accuracy. Hindsight and the profile each had 75% accuracy. All queries were answered, with 100% coverage and no schema failure. Full history produced two confident

wrong event-level predictions; other arms produced none. Because vectors recur across test events, these counts are not independent generation counts.

Table 1: Scores on 128 held-out outcomes from 16 dogs. Lower Brier and log loss are better; ECE is descriptive.

Evidence arm	Brier	Accuracy (%)	Log loss	ECE	Confident wrong
No history	0.5532	50.8	0.8364	0.1047	0
Full history	0.3806	76.6	0.6319	0.1071	2
Profile	0.3716	75.0	0.6208	0.1006	0
Vector retrieval	0.4339	68.0	0.7038	0.0820	0
Hindsight	0.3761	75.0	0.6281	0.1028	0

Table 2: Paired Brier improvement, comparator minus Hindsight. Positive values favor Hindsight. Intervals condition on fixed generations.

Comparator	Δ	Dog 95% interval	Site 95% interval
No history	0.1771	[0.0774, 0.2756]	[0.0677, 0.2627]
Full history	0.0046	[−0.0087, 0.0169]	[−0.0091, 0.0157]
Profile	−0.0044	[−0.0111, 0.0023]	[−0.0117, 0.0031]
Vector retrieval	0.0578	[0.0118, 0.1067]	[0.0046, 0.1081]

Hindsight substantially improved on no individual history, but that contrast does not establish a benefit unique to structured memory. Its mean advantage over full history was 0.0046, with the interval spanning zero; its mean disadvantage versus the profile was 0.0044. The upper 95% bounds for improvement over both simpler methods were below 0.03 under this conditional bootstrap. These results do not support the specified practical gain; they are not a general equivalence test across models or settings.

Hindsight’s improvement over top-three retrieval had positive dog and site 95% intervals, but their lower bounds (0.0118 and 0.0046) were below 0.03. The one-sided adjusted lower bounds against full history, profile, and retrieval were respectively −0.0099, −0.0116, and 0.0080 for dogs, and −0.0103, −0.0122, and 0.0009 for sites. Thus practical superiority over all history baselines failed. Appendix B includes every dog-level mean, including unfavorable cases.

5.2 Separate 420-dog diagnostic

The pooled pointing-side baseline scored 0.5180 Brier, with 51.4% accuracy. The primary point-relative-history model scored 0.5375: paired improvement −0.01944, dog interval [−0.02849, −0.01067], and site interval [−0.02862, −0.01093]. This primary finding is unfavorable to that historical representation. Physical-side history with pooled shrinkage was a secondary analysis and scored 0.4103, improvement 0.10772, dog interval [0.09101, 0.12438], and site interval [0.08712, 0.12967]. Its accuracy was 69.7%. The uniformly smoothed physical-side model scored 0.4154. The data included 66 training and 63 test no-choice outcomes. Table 3 preserves all five diagnostic scores.

Pooled physical-side accuracy is unusually low because pooled training left and right counts tie exactly. Leaving out a dog’s own history slightly shifts the remaining pool against that dog’s side bias, and argmax exaggerates this tiny probability difference. Its Brier score is close to the pooled

Table 3: Separate frequency-model diagnostic: 420 dogs, 3,360 held-out attempts, 20 sites. These are not Hindsight arms.

Frequency predictor	Brier	Accuracy (%)	Log loss
Pooled physical side	0.5190	27.6	0.7742
Pooled pointing side (comparator)	0.5180	51.4	0.7736
Physical side, uniform shrinkage	0.4154	69.7	0.7125
Physical side, pooled shrinkage	0.4103	69.7	0.6582
Point-relative history (primary)	0.5375	52.1	0.8293

pointing comparator. This tie-sensitive accuracy artifact is not evidence of a large probabilistic failure. A labeled post hoc sensitivity using inclusive pooling retained the direction of the primary and shrinkage contrasts.

5.3 Fidelity, grounding, and reconstruction audits

An independent training-data audit recovered all 128 distinct supplied observations in both Hindsight recalls per dog: 256/256 observation–query pairs had correct core human-point/dog-choice triplets, without omitted or contradictory core trials. However, recall contained an invented event date and an unsupported causal interpretation about choosing by position. Ingestion timestamps appeared despite unknown behavioral dates. Correct recovery of positions therefore does not establish complete semantic fidelity or reliable temporal provenance.

A separate outcome-blind qualitative review of all 132 unique rationales found no definite incorrect historical claim or factual observation lacking packet support. Two wordings were ambiguous: a profile rationale conflated pointed sides with cue conditions, and a Hindsight rationale called nonchoice risk “refusal.” General assumptions and predictive hypotheses were distinguished from asserted observations. Arm identities were visible, and these were qualitative judgments rather than independently blinded psychological assessments. Hindsight rationales did not repeat the invented event dates. No causal-explanation validity claim follows.

The independent scoring audit used code that did not import the scorer. It checked all 65 prediction hashes and plan/scorer hashes, matched the 128 query/outcome records to source data, and reconstructed the metrics and 10,000 paired dog/site resamples to within 10^{-12} . This checks current artifact consistency and numerical implementation; it cannot certify that no actor ever accessed an outcome or that the model had not encountered public data during training.

5.4 Resources

Corrected Hindsight retention took 36.9 minutes of measured elapsed retention time on the local CPU (median 137.1 seconds per dog), with 56,284 locally reported extraction tokens. These are wall-clock measurements, not process CPU time. Median recall was 0.215 seconds per query; conventional retrieval with a warm embedder took 0.030 seconds. These measurements exclude setup and predictor inference, differ in initialization conditions, and are not controlled deployment benchmarks. Discarded extraction and development consumed additional compute. No paid API, cloud memory service, or additional monetary purchase was used. Codex predictions consumed an existing account allowance; hardware, electricity, and internet access remain economic resources. “Zero additional spending” is not zero total cost.

6 Discussion

The narrow empirical finding is that this corrected local Hindsight retain/recall pipeline did not demonstrate a practically meaningful Brier advantage over either complete supplied history or a deterministic profile. The profile’s point estimate was best, but the present design does not establish universal superiority of the profile. A cautious operational decision is to keep simple history methods as reference implementations and require a larger, prospective comparison before adding memory-system complexity.

The retrieval contrast also requires care. Hindsight effectively supplied all eight historical trials; cosine top-three supplied three. Its advantage may reflect available evidence coverage, extraction, presentation, or model variability rather than a unique architectural mechanism. With the entire historical corpus fitting easily into context, this is a weak stress test of long-term memory, interference, consolidation, and scalable search. Stronger future comparisons should match evidence budgets where relevant, vary corpus size, include full-history feasibility limits, and separately ablate retrieval coverage and generative transformation.

The larger diagnostic shows why “history helps” is underspecified. Point-relative frequency history worsened its designated primary score while physical-side shrinkage improved a secondary score. Representation matters, and a favorable representation result cannot retrospectively rescue an unfavorable primary one. Those descriptive findings neither prove a stable behavioral trait nor attribute improvement to Hindsight.

Several threats constrain interpretation. Sixteen deterministically selected dogs and 12 represented sites provide limited population precision, while eight test attempts per dog cannot establish benefit for a specified individual. Selection on included sessions and observed first-eight labels changes the target population. Omitted intervening attempts, reinforcement, order effects, laboratory handling, and other unobserved experience can affect later choices. The second pointing condition is an experimental context change, not verified lifetime novelty. Public source data and published conclusions may have appeared in language-model training; anonymized packets and no browsing reduce direct lookup without excluding contamination.

Only one stochastic output per distinct input was retained. Exact predictor revision and sampling parameters are unavailable, so the recorded vectors allow score reconstruction but not guaranteed generational reproduction. Cluster intervals do not include model-generation variance or extraction-model alternatives. The training-only amendment was justified by source fidelity, yet occurred after some baseline generation and limits confirmatory status. The small local extraction model, disabled consolidation, RRF setting, and absence of reflect further restrict conclusions to this implementation. Instruction-only file isolation, qualitative nonblind grounding review, and temporal metadata errors remain material methodological qualifications.

A future test of the original everyday single-dog question requires genuinely future observations, an operational response codebook fixed before collection, many independently sampled cues, a frozen memory comparison, and prospective recording of both failures and nonresponses. Researcher-edited or social-media-selected clips alone cannot supply that evidence. Prospective sample size should target the chosen practical difference while accounting for correlated events, contexts, and repeated stochastic predictions. This paper offers a bounded retrospective result, not a completed prospective animal-behavior validation.

7 Conclusion

In a 16-dog retrospective laboratory proxy, Hindsight achieved a Brier score of 0.3761 but did not demonstrate the specified practical improvement over full supplied history (0.3806) or a deterministic profile (0.3716). It outperformed the chosen top-three retrieval baseline in conditional interval estimates without clearing the practical lower-bound criterion. The original claim about one dog’s prospective everyday responses remains unestablished. The appropriate conclusion is configuration- and endpoint-specific, with simple historical representations retained as strong controls.

Data, code, and research transparency

The ManyDogs trial data and codebook are publicly available under the source repository’s CC BY 4.0 license [3]; contributors retain credit. Derived analyses use anonymized selection and chronological splitting. The local accompanying study bundle contains source CSVs, exclusion records, plans, versioned memory evidence, prompts, all probability vectors, locks, scoring code, independent reconstruction, qualitative audits, and runtime manifests. Relative artifact paths appear in Appendix D. No public archive or DOI has been deposited for this derived bundle; public release and an archival identifier remain tasks for a human author before submission. The standalone manuscript itself uses no external images, bibliography file, or project assets.

No new animal intervention or data collection was conducted for this analysis. We rely on the source study’s reporting of its original procedures; we do not claim a new ethics approval. No author identity, affiliation, funding award, external registration, peer review, or arXiv acceptance is asserted. This draft was developed through an AI-assisted multi-agent workflow with a supervising agent, an orchestration agent, and specialist methodological and source reviews. Human authors must determine authorship, verify the final manuscript, state applicable disclosures and conflicts, and take responsibility before any submission. The draft has not been submitted to arXiv.

A Eligibility audit

Exclusion reasons overlap and therefore do not sum to 284: inconsistent/nonincluded status, 249; fewer than eight nonostensive attempts, 172; fewer than eight ostensive attempts, 149; unknown label among the first eight nonostensive attempts, 123; analogous ostensive unknown label, 123; experimental source-order mismatch, 36; invalid or inconsistent condition order, three; nonsequential nonostensive trial numbering, three; and nonsequential ostensive numbering, one. The eligible diagnostic cohort included 212 dogs with ostensive-first and 208 with nonostensive-first order. The corresponding primary point-relative improvements were -0.01469 and -0.02428 ; these strata are descriptive.

B All dog-level Brier scores

Each row averages eight held-out attempts. These are descriptive paired outcomes, not sufficiently precise individual-dog treatment estimates. Source site codes are shown only after prediction lock.

Dog	Site	No history	Full	Profile	Retrieval	Hindsight
D01	queensu	0.5650	0.1292	0.1202	0.2381	0.1208
D02	duke	0.6418	0.6182	0.6046	0.6381	0.6100

Dog	Site	No history	Full	Profile	Retrieval	Hindsight
D03	accc	0.5650	0.6244	0.5949	0.6078	0.6050
D04	ucs	0.5218	0.3918	0.3918	0.3904	0.3986
D05	eltebuda	0.3900	0.3374	0.3209	0.2934	0.3342
D06	dcc	0.5650	0.5490	0.5373	0.5130	0.5460
D07	bccc	0.5218	0.2863	0.2314	0.5296	0.2248
D08	cci	0.7775	0.7014	0.6868	0.6434	0.6964
D09	auburn	0.6525	0.4016	0.3672	0.4743	0.3836
D10	eltebuda	0.5218	0.0896	0.0986	0.1572	0.0960
D11	duke	0.5218	0.1418	0.1371	0.3560	0.1660
D12	umessina	0.3900	0.4616	0.4683	0.5848	0.4578
D13	manitoba	0.5218	0.3722	0.3872	0.4612	0.3626
D14	bccc	0.5650	0.1054	0.1054	0.2108	0.1076
D15	bdl	0.5650	0.7015	0.7399	0.6536	0.7661
D16	manitoba	0.5650	0.1784	0.1545	0.1900	0.1414

C Runtime and reproducibility details

The local stack used Hindsight API/client 0.10.2, Ollama 0.32.5, and isolated Python 3.12.13, with CPU inference and no GPU. The extraction tag `qwen3.5:4b` was a cached Q4_K_M model; the runtime manifest reports 4.7B parameters. Its SHA256 digest is

2a654d98e6fba55d452b7043684e9b57a947e393bbffa62485a7aac05ee4eefd.

Thinking was disabled, context capped at 8,192 tokens, output capped at 2,048 tokens, and LLM concurrency set to one. The embedder revision is

614241f622f53c4eeff9890bdc4f31cfecc418b3;

the local ONNX SHA256 is

ca456c06b3a9505ddfd9131408916dd79290368331e7d76bb621f1cba6bc8665.

The source main CSV SHA256 is

f1b497e835b10b7b5fc030acc2e98d90974e37422c49e92851440e2d87e2f02f.

Model artifacts and checkpoints are preserved locally. These versions identify extraction and embedding components, not the unexposed Codex predictor revision.

D Artifact map and reconstruction

Paths below are relative to the accompanying `public_dog_pilot` directory, not public-download links. They permit reconstruction from saved outputs without repeating model calls.

Source and selection

`data/md1_data.csv`, `data/md1_data_pilot.csv`; `secondary_packets/selection_manifest.json`; `results_statistical/exclusions.csv`; `secondary_packets/omitted_intervening_attempts.json`.

Plans and prompts

prereg_llm.md, prereg_statistical.md; prediction_packets/common_instruction.txt; versioned arm packets in prediction_packets/.

Probabilities and prediction lock

predictions/; results_llm/prediction_lock.json; the lock lists all 65 output hashes.

Scoring and reconstruction

score_llm.py, audit_llm.py; results_llm/summary.json, results_llm/per_dog.csv, results_llm/independent_audit.json; statistical_pilot.py and results_statistical/.

Memory and grounding

runtime/build_memory_evidence_v2.py; secondary_packets/hindsight_evidence_v2.json, secondary_packets/vector_rag_evidence_v2.json; extraction_audit_initial.md, extraction_audit_v2.md, rationale_audit.md and their structured audit records.

Versions and resource accounting

runtime/model_manifest.json, runtime/requirements.lock.txt, runtime/retention_v2_lock.json, runtime/retained_banks_v2.json; saved recall and vector latency records.

Running the scorer revalidates and hashes predictions before loading labels; independent audit code reconstructs rather than imports the scorer. For publication, a human author should release a licensed artifact snapshot with a stable version and checksum, and document exact predictor settings if recoverable. Generational reproduction remains limited until those settings are known.

References

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